

# CS & IT ENGINEERING



## COMPUTER ORGANIZATION AND ARCHITECTURE

Pipeline Processing

Lecture No.- 5



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# Recap of Previous Lecture



*Doubt ?*



Topic

Data Hazards

Topic

Operand Forwarding

Topic

Branch Prediction



# Topics to be Covered



Topic

Data Hazard Classification

Topic

RAW, WAW, WAR

Topic

CPI Calculation

#Q. The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is 219?

|    |   |    |    |    |
|----|---|----|----|----|
| IF | 1 |    |    |    |
| ID | 1 |    |    |    |
| OF | 1 | 40 | 35 | 25 |
| PO |   | 3  | 2  | 1  |
| WB | 1 |    |    |    |

$$\begin{aligned}
 \text{stalls} &= 40 * 2 & 35 * 1 \\
 &= 80 & = 35 \\
 &\underbrace{\hspace{10em}} \\
 &\text{Total} = 115
 \end{aligned}$$

$$\begin{aligned}
 \text{w/o hazard} &= 5 + 100 - 1 = 104 \\
 \text{stalls} &= 115 \\
 \hline
 \text{Total} &= 219
 \end{aligned}$$



[NAT]



PYQ (GATE)

#Q. Consider a 4-stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below:

|    | S <sub>1</sub> | S <sub>2</sub> | S <sub>3</sub> | S <sub>4</sub> |
|----|----------------|----------------|----------------|----------------|
| I1 | 2              | 1              | 1              | 1              |
| I2 | 1              | 3              | 2              | 2              |
| I3 | 2              | 1              | 1              | 3              |
| I4 | 1              | 2              | 2              | 2              |

Ans = 23

What is the number of cycles needed to execute the following loop?

For (i = 1 to 2) { I1; I2; I3; I4 }

[illegible]

## [Solution]



|    | $S_1$ | $S_2$ | $S_3$ | $S_4$ |
|----|-------|-------|-------|-------|
| I1 | 2     | 1     | 1     | 1     |
| I2 | 1     | 3     | 2     | 2     |
| I3 | 2     | 1     | 1     | 3     |
| I4 | 1     | 2     | 2     | 2     |

|    | S1 | S2 | S3 | S4 |
|----|----|----|----|----|
| I1 | 2  | 3  | 4  | 5  |
| I2 | 3  | 6  | 8  | 10 |
| I3 | 5  | 7  | 9  | 13 |
| I4 | 6  | 9  | 11 | 15 |
| I1 | 8  | 10 | 12 | 16 |
| I2 | 9  | 13 | 15 | 18 |
| I3 | 11 | 14 | 16 | 21 |
| I4 | 12 | 16 | 18 | 23 |



ques) from prev. quest<sup>n</sup>, no. of cycles needed for

loop  $i = 1$     I2    S4     $\Rightarrow$  10

$i = 2$     I3    S4     $\Rightarrow$  11

$i = 2$     I1    S4     $\Rightarrow$  16



Data



## Topic : Hazard Classification

Assume that there are two instructions  $i$  and  $j$ , and  $i$  is executed before  $j$

→ RAW

→ WAW

→ WAR





## Topic : Read After Write (RAW)

→ True data dependency

$j$  tries to read a source before  $i$  writes it. So  $j$  incorrectly gets the old value.

$i: R1 \leftarrow R2 + R3$   
 $j: R6 \leftarrow R1 * R5$  } so,  $\Rightarrow$  operand forwarding





→ false dependency  
→ write dependency

$i: R1 \leftarrow R2 + R3$   
 $j: R1 \leftarrow R4 * R5$

$\} \underline{\underline{\text{Sol}^n}} :- \text{Register Renaming}$   
 $\Downarrow$

$i: R1 \leftarrow R2 + R3$   
 $j: R8 \leftarrow R4 * R5$

$$j: R8 \leftarrow R4 * R5$$



## Topic : Write After Read (WAR)

→ False dependency  
→ Anti-dependency

$j$  tries to write an operand before  $i$  reads it, hence  $i$  incorrectly gets new value

$i: R1 \leftarrow R2 + R3$   
 $j: R2 \leftarrow R4 * R5$  } sol<sup>n</sup>  $\Rightarrow$  Register Renaming  
 $\Downarrow$   
 $R1 \leftarrow R2 + R3$   
 $R7 \leftarrow R4 * R5$

WAW & WAR  $\Rightarrow$  no stalls ever

Register Renaming  $\Rightarrow$  Hardware solution



#Q. Count the number of RAW, WAW and WAR dependencies?

|    |                |                         |
|----|----------------|-------------------------|
| 1. | ADD R2, R1, R0 | $R2 \leftarrow R1 + R0$ |
| 2. | MUL R4, R3, R2 | $R4 \leftarrow R3 * R2$ |
| 3. | SUB R6, R5, R4 | $R6 \leftarrow R5 - R4$ |
| 4. | ADD R6, R7, R8 | $R6 \leftarrow R7 + R8$ |
| 5. | MUL R7, R1, R2 | $R7 \leftarrow R1 * R2$ |
| 6. | SUB R1, R3, R4 | $R1 \leftarrow R3 - R4$ |

#Q. An instruction format has the following structure

Instruction number: opcode destination reg, source reg-1, source reg-2

Consider the following sequence of instruction to be executed in a pipelined processor

|                    |                         |
|--------------------|-------------------------|
| I1: DIV R3, R1, R2 | $R3 \leftarrow R1 / R2$ |
| I2: SUB R5, R3, R4 | $R5 \leftarrow R3 - R4$ |
| I3: ADD R3, R5, R6 | $R3 \leftarrow R5 + R6$ |
| I4: MUL R7, R3, R8 | $R7 \leftarrow R3 * R8$ |

Which of the following statement is/are TRUE?

- ☐ A There is a WAW dependency on R3 between I3 and I4
- ☒ B There is a WAW dependency on R3 between I1 and I3
- ☒ C There is a RAW dependency on R3 between I1 and I2
- ☐ D There is a RAW dependency on R3 between I2 and I3



## [MSQ]



#Q. For a pipelined CPU with a single ALU, consider the following situations

- I. The  $J + 1^{\text{st}}$  instruction uses the result of the  $J^{\text{th}}$  instruction as an operand  $\rightarrow$  data hazard
- II. The execution of a conditional jump instruction  $\rightarrow$  branch hazard
- III. The  $J^{\text{th}}$  and  $J + 1^{\text{st}}$  instructions require the ALU at the same time  $\rightarrow$  structural hazard

Which of the above can cause a hazard

- A. I and II only
- B. II and III only
- C. III only
- D. ☒ All the three



#Q. Register renaming is done in pipelined processors:

- a) as an alternative to register allocation at compile time
- b) for efficient access to function parameters and local variables
- ☒ c) to handle certain kinds of hazards
- d) as part of address translation



#Q. Consider the following sequence of instructions, which is to be executed on a 5-stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding
2. When pipeline does not use operand forwarding

ADD R2, R1, R0

$R2 \leftarrow R1 + R0$

MUL R4, R3, R2

$R4 \leftarrow R3 * R2$

SUB R6, R5, R4

$R6 \leftarrow R5 - R4$

ADD R6, R7, R8

$R6 \leftarrow R7 + R8$

MUL R7, R1, R2

$R7 \leftarrow R1 * R2$

SUB R1, R3, R4

$R1 \leftarrow R3 - R4$



#Q. Consider the following sequence of instructions, which is to be executed on a 5-stage pipeline: IF, ID, OF, EX and WB. Each stage takes one cycle for each instruction. Total Cycles required to execute this sequence

1. When pipeline uses operand forwarding
2. When pipeline does not use operand forwarding

|                 |                         |
|-----------------|-------------------------|
| ADD R2, R1, R0  | $R2 \leftarrow R1 + R0$ |
| MUL R4, R3, R2  | $R4 \leftarrow R3 * R2$ |
| LOAD R7, (1000) | $R7 \leftarrow M[1000]$ |
| ADD R6, R7, R8  | $R6 \leftarrow R7 + R8$ |
| MUL R7, R1, R2  | $R7 \leftarrow R1 * R2$ |
| SUB R1, R3, R4  | $R1 \leftarrow R3 - R4$ |





## Topic : CPI Calculation in Pipeline

without hazard :-

for  $n$  no. of inst<sup>ns</sup>, no. of cycles =  $k + n - 1$

$$\text{for } 1 \text{ ———— } = \boxed{\frac{k + n - 1}{n}} \text{ CPI}$$

In ideal condition  $\Rightarrow$

$$\boxed{\text{CPI} = 1}$$

with hazard:-

assume total stalls no. of stalls =  $x$

for  $n$  inst<sup>ns</sup>, no. of cycles =  $k+n-1+x$

for 1 —|| ————— 
$$CPI = \frac{k+n-1+x}{n}$$

In ideal condition:-

$$CPI = \frac{n+x}{n}$$

$$CPI = 1 + \frac{x}{n}$$

avg. inst<sup>n</sup> execution time = CPI<sub>avg</sub> \* cycle time

$$S_{ideal} \text{ (without hazard)} = \frac{t_n}{t_p}$$

$$S_{ideal} \text{ (with hazard)} = \frac{t_n}{CPI_{avg} * t_p}$$



20% of inst<sup>ns</sup>

#Q. Consider a 5-stage pipeline which is executing a program of 1000 instructions.

Among all instructions 200 instructions cause 2 stall cycles each.

Total stalls (x) =  $200 * 2 = 400$

1. Calculate CPI of pipeline?  $\frac{5 + 1000 - 1 + 400}{1000} = 1.404$
2. If pipeline cycle time is 3ns then what is average instruction execution time?  $1.404 * 3 = 4.212 \text{ ns}$
3. Calculate CPI of pipeline in ideal conditions?  $1 + \frac{400}{1000} = 1.4$
4. If pipeline cycle time is 3ns then what is average instruction execution time in ideal conditions?  $1.4 * 3 = 4.2 \text{ ns}$

CPI:-

in ideal condition

$$CPI = 1 + \frac{400}{1000}$$

$$= 1 + \frac{200 * 2}{1000}$$

$$= 1 + 0.2 * 2$$

$$= 1.4$$

Ques:- Ideal condition

20% inst<sup>ns</sup> cause 2 stalls each

$$CPI = ?$$

Sol<sup>n</sup>

$$CPI = 1 + 0.2 * 2$$
$$= 1.4$$

ans)

30% inst<sup>ns</sup> cause 1 stall each

$$\begin{aligned} CPI_{ideal} &= 1 + 0.3 * 1 \\ &= 1.3 \end{aligned}$$



#Q. Consider a 5-stage instruction pipeline, where all stages take equal delay. <sup>2ns.</sup> When an application is executing on this 5-stage pipeline, consider 20% of the instructions incur 3 pipeline stall cycles is.

1. Average CPI for pipeline?  $= 1 + 0.2 * 3 = 1.6$
2. Average instruction execution time for pipeline?  $1.6 * 2ns = 3.2 ns.$
3. The speedup of pipeline achieved with respect to non-pipeline?  $\Rightarrow 3.125$

$$t_p = \max(2, 2, 2, 2, 2) = 2ns$$

$$t_n = 2 + 2 + 2 + 2 + 2 = 10ns$$

$$S = \frac{10ns}{1.6 * 2} = 3.125$$

#Q. Consider a 5-stage instruction pipeline, where stages take delays 5ns, 4ns, 6ns, 4ns and 5ns respectively. The pipeline is used to execute a program in which 25% instructions cause 4 stalls due to hazard. The average instruction execution time in the pipeline is 12 ns?

$$t_p = \max(5, 4, 6, 4, 5) = 6 \text{ ns}$$

$$\text{CPI}_{\text{avg}} = 1 + 0.25 * 4$$

$$= 2$$

$$\text{A.I.E.T.} = 6 * 2 = 12 \text{ ns}$$

Speed up of pipeline as compared to non-pipeline 2?

$$\text{sol}^n$$

$$t_n = 5 + 4 + 6 + 4 + 5 = 24 \text{ ns}$$

$$S = \frac{24}{6 * 2} = 2$$



[NAT]

GATE-PYQ

$$t_p = \frac{1}{1\text{GHz}} = 1\text{ns}$$



#Q. A CPU has a five-stage pipeline and runs at 1 GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline. The processor stops fetching new instructions following a conditional branch until the branch outcome is known. A program executes  $10^9$  instructions out of which 20% are conditional branches. If each instruction takes one cycle to complete on average, the total execution time of the program is 1.4 seconds?

→ stalls due to each branch inst<sup>n</sup> =  $3 - 1$   
= 2

$$CPI_{avg} = 1 + 0.2 * 2 = 1.4$$

$$AET = 1.4 * 1\text{ns} = 1.4\text{ns}$$

$$\text{Prog. execut<sup>n</sup> time} = 1.4\text{ns} * 10^9 = 1.4\text{sec.}$$



Ques) 20% inst<sup>ns</sup> cause 2 stalls due to branch hazard  
15% — 11 — 3 — 11 — data hazard

CPI<sub>avg</sub> = \_\_\_\_\_ ?

sol<sup>n</sup>

$$\begin{aligned} \text{CPI}_{\text{avg}} &= 1 + (0.2 * 2) + (0.15 * 3) \\ &= 1.85 \end{aligned}$$

Ques) 20% branch inst<sup>ns</sup>

10% data dependent inst<sup>ns</sup>

branch condition evaluated in 5<sup>th</sup> stage  $\Rightarrow$  stalls = 5 - 1 = 4  
each data hazard causes 2 stalls.

$CPI_{avg} = \text{---} ?$

Sol<sup>n</sup>

$$CPI_{avg} = 1 + (0.2 * 4) + (0.1 * 2) \\ = 2$$

Ques) No branch hazard and no any structural hazard.

20% data dependent inst<sup>ns</sup>.

30% of <sup>all</sup> data dependent inst<sup>ns</sup> cause 2 stalls each  
and remaining no stalls because of operand forwarding.

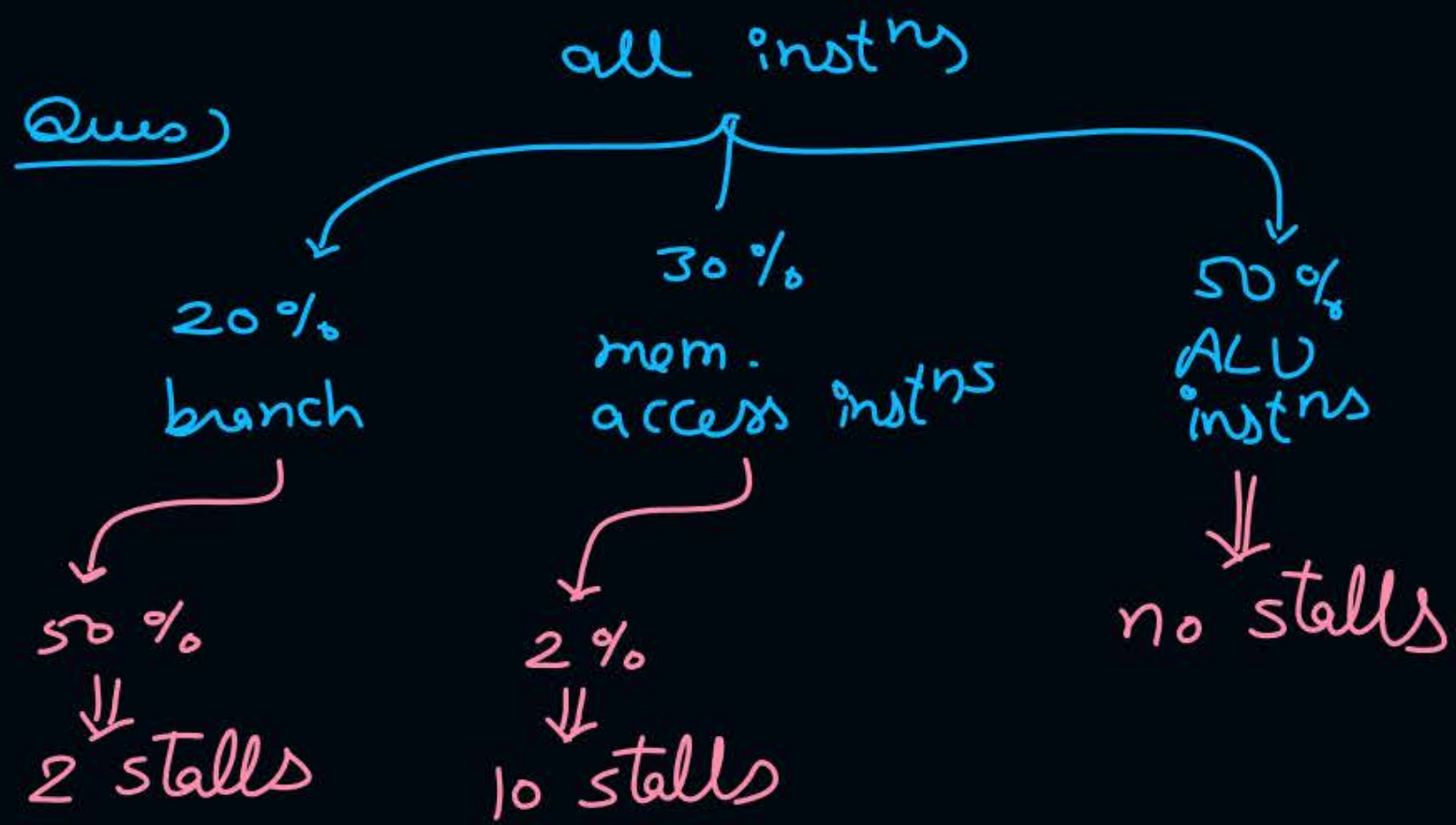
$$CPI_{avg} = \text{---} ?$$

Sol<sup>n</sup>

all  
20% data hazard  
30%  
↓ 2 stalls

$$CPI_{avg} = 1 + 0.2 * 0.3 * 2$$
$$= 1.12$$





$$CPI_{avg} = 1 + (0.2 * 0.5 * 2) + (0.3 * 0.02 * 10)$$
$$= 1.26$$



## 2 mins Summary



**Topic**

Data Hazard Classification

**Topic**

RAW, WAW, WAR

**Topic**

CPI Calculation





**Happy Learning**

**THANK - YOU**

